



17.0
BE--Basic Edition

Statistics and Data Science

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Notes:

1. Unicode is supported; see [help unicode advice](#).

```
1 . do "C:\Users\fahim\AppData\Local\Temp\STD4280_000000.tmp"
2 .
3 . /* Elastance Statistics Test*/
4 .
5 . clear all

6 . import excel "C:\Users\fahim\Downloads\PV Loop Statistics-Raw Data.xlsx", sheet("FINAL-ESPVR") firstrow
   (12 vars, 45 obs)

7 . destring Ees V0, replace force
   Ees already numeric; no replace
   V0 already numeric; no replace

8 .
9 .
10 . xtset Number TimeNumerical
```

Panel variable: **Number** (unbalanced)
Time variable: **TimeNumerical**, 0 to 74, but with gaps
Delta: 1 unit

```
11 . xtmixed Ees GroupNumerical##TimeNumerical Sex weight || Number:, var reml
```

Performing EM optimization:

Performing gradient-based optimization:

```
Iteration 0:  log restricted-likelihood = -57.32187
Iteration 1:  log restricted-likelihood = -57.32186
Iteration 2:  log restricted-likelihood = -57.32186
```

Computing standard errors:

Mixed-effects REML regression	Number of obs	=	45
Group variable: Number	Number of groups	=	16
	Obs per group:		
	min	=	2
	avg	=	2.8
	max	=	3
	Wald chi2(10)	=	105.97
Log restricted-likelihood = -57.32186	Prob > chi2	=	0.0000

Ees	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
GroupNumerical						
1	-.2237883	.6936344	-0.32	0.747	-1.583287	1.13571
2	.2544525	.6312105	0.40	0.687	-.9826974	1.491602
TimeNumerical						
30	1.213473	.4659702	2.60	0.009	.3001881	2.126758
74	.3368063	.4659702	0.72	0.470	-.5764785	1.250091
GroupNumerical#TimeNumerical						
1 30	.124027	.7099209	0.17	0.861	-1.267392	1.515446
1 74	1.069637	.7525261	1.42	0.155	-.4052874	2.544561
2 30	2.546527	.6390367	3.98	0.000	1.294038	3.799016
2 74	1.655116	.6597445	2.51	0.012	.3620404	2.948191
Sex	.5894072	.4629207	1.27	0.203	-.3179006	1.496715
weight	.0163868	.0399771	0.41	0.682	-.061967	.0947406
_cons	.0688342	3.164573	0.02	0.983	-6.133614	6.271283

Random-effects parameters	Estimate	Std. err.	[95% conf. interval]	
Number: Identity				
var(_cons)	.497887	.3057902	.1493975	1.659275
var(Residual)	.5737189	.1685748	.3225464	1.020484

LR test vs. linear model: $\text{chibar2}(01) = 6.35$ Prob >= chibar2 = 0.0059

12 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
Ees			
TimeNumerical@GroupNumerical			
0	2	7.57	0.0227
1	2	8.19	0.0167
2	2	74.00	0.0000
Joint	6	89.74	0.0000

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
Ees						
TimeNumerical@GroupNumerical						
(30 vs base) 0	1.213473	.4659702	2.60	0.009	.3001881	2.126758
(30 vs base) 1	1.3375	.5355926	2.50	0.013	.2877577	2.387242
(30 vs base) 2	3.76	.4373095	8.60	0.000	2.902889	4.617111
(74 vs base) 0	.3368063	.4659702	0.72	0.470	-.5764785	1.250091
(74 vs base) 1	1.406443	.592285	2.37	0.018	.2455857	2.5673
(74 vs base) 2	1.991922	.4657898	4.28	0.000	1.078991	2.904853

13 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
Ees			
GroupNumerical@TimeNumerical			
0	2	0.49	0.7808
30	2	26.35	0.0000
74	2	9.01	0.0110
Joint	6	34.65	0.0000

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
Ees						
GroupNumerical@TimeNumerical						
(1 vs base) 0	-.2237883	.6936344	-0.32	0.747	-1.583287	1.13571
(1 vs base) 30	-.0997613	.6722956	-0.15	0.882	-1.417436	1.217914
(1 vs base) 74	.8458484	.7210986	1.17	0.241	-.5674789	2.259176
(2 vs base) 0	.2544525	.6312105	0.40	0.687	-.9826974	1.491602
(2 vs base) 30	2.80098	.6138237	4.56	0.000	1.597907	4.004052
(2 vs base) 74	1.909568	.6361579	3.00	0.003	.6627218	3.156415

14 .
 15 .
 16 .
 17 . xtset Number TimeNumerical

Panel variable: **Number** (unbalanced)
 Time variable: **TimeNumerical**, 0 to 74, but with gaps
 Delta: 1 unit

18 . xtmixed Ees Sex weight i.GroupNumerical i.TimeNumerical || Number:, var reml

Performing EM optimization:

Performing gradient-based optimization:

Iteration 0: log restricted-likelihood = **-67.869859**
 Iteration 1: log restricted-likelihood = **-67.868869**
 Iteration 2: log restricted-likelihood = **-67.868869**

Computing standard errors:

Mixed-effects REML regression	Number of obs	=	45
Group variable: Number	Number of groups	=	16
	Obs per group:		
	min	=	2
	avg	=	2.8
	max	=	3
	Wald chi2(6)	=	57.45
Log restricted-likelihood = -67.868869	Prob > chi2	=	0.0000

Ees	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
Sex	.5579913	.4662894	1.20	0.231	-.3559191	1.471902
weight	.0185747	.0399793	0.46	0.642	-.0597832	.0969327
GroupNumerical						
1	.1956932	.5551965	0.35	0.724	-.8924719	1.283858
2	1.709996	.5050156	3.39	0.001	.7201836	2.699808
TimeNumerical						
30	2.256959	.3520772	6.41	0.000	1.5669	2.947018
74	1.270103	.3675137	3.46	0.001	.5497898	1.990417
_cons	-.8036061	3.155756	-0.25	0.799	-6.988774	5.381561

Random-effects parameters	Estimate	Std. err.	[95% conf. interval]	
Number: Identity				
var(_cons)	.3686061	.3136524	.0695444	1.953721
var(Residual)	.9506188	.2575838	.5589347	1.616783

LR test vs. linear model: $\text{chibar2}(01) = 2.39$ Prob >= chibar2 = 0.0609

19 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
Ees			
TimeNumerical@GroupNumerical			
0	2	41.17	0.0000
1	2	41.17	0.0000
2	2	41.17	0.0000
Joint	2	41.17	0.0000

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
Ees						
TimeNumerical@GroupNumerical						
(30 vs base) 0	2.256959	.3520772	6.41	0.000	1.5669	2.947018
(30 vs base) 1	2.256959	.3520772	6.41	0.000	1.5669	2.947018
(30 vs base) 2	2.256959	.3520772	6.41	0.000	1.5669	2.947018
(74 vs base) 0	1.270103	.3675137	3.46	0.001	.5497898	1.990417
(74 vs base) 1	1.270103	.3675137	3.46	0.001	.5497898	1.990417
(74 vs base) 2	1.270103	.3675137	3.46	0.001	.5497898	1.990417

20 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
Ees			
GroupNumerical@TimeNumerical			
0	2	12.81	0.0017
30	2	12.81	0.0017
74	2	12.81	0.0017
Joint	2	12.81	0.0017

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
Ees						
GroupNumerical@TimeNumerical						
(1 vs base) 0	.1956932	.5551965	0.35	0.724	-.8924719	1.283858
(1 vs base) 30	.1956932	.5551965	0.35	0.724	-.8924719	1.283858
(1 vs base) 74	.1956932	.5551965	0.35	0.724	-.8924719	1.283858
(2 vs base) 0	1.709996	.5050156	3.39	0.001	.7201836	2.699808
(2 vs base) 30	1.709996	.5050156	3.39	0.001	.7201836	2.699808
(2 vs base) 74	1.709996	.5050156	3.39	0.001	.7201836	2.699808

21 .
 22 .
 23 .
 24 . xtset Number TimeNumerical

Panel variable: **Number** (unbalanced)
 Time variable: **TimeNumerical**, 0 to 74, but with gaps
 Delta: 1 unit

25 . xtmixed V0 GroupNumerical##TimeNumerical Sex weight || Number:, var reml

Performing EM optimization:

Performing gradient-based optimization:

```
Iteration 0:  log restricted-likelihood = 102.21945
Iteration 1:  log restricted-likelihood = 704.04929 (not concave)
Iteration 2:  log restricted-likelihood = 704.06115 (not concave)
Iteration 3:  log restricted-likelihood = 704.06627 (not concave)
Iteration 4:  log restricted-likelihood = 704.06628 (not concave)
Iteration 5:  log restricted-likelihood = 704.06629 (not concave)
Iteration 6:  log restricted-likelihood = 704.06629 (not concave)
Iteration 7:  log restricted-likelihood = 704.06629 (not concave)
Iteration 8:  log restricted-likelihood = 704.06629 (not concave)
Iteration 9:  log restricted-likelihood = 704.06629 (not concave)
Iteration 10: log restricted-likelihood = 704.06629 (not concave)
Iteration 11: log restricted-likelihood = 704.06629 (not concave)
Iteration 12: log restricted-likelihood = 704.06629 (not concave)
Iteration 13: log restricted-likelihood = 704.06629 (not concave)
Iteration 14: log restricted-likelihood = 704.06629 (not concave)
Iteration 15: log restricted-likelihood = 704.06629 (not concave)
Iteration 16: log restricted-likelihood = 704.06629 (not concave)
Iteration 17: log restricted-likelihood = 704.06629 (not concave)
```

[illegible]

[illegible]

[illegible]

[illegible]

```

Iteration 270: log restricted-likelihood = 704.06629 (not concave)
Iteration 271: log restricted-likelihood = 704.06629 (not concave)
Iteration 272: log restricted-likelihood = 704.06629 (not concave)
Iteration 273: log restricted-likelihood = 704.06629 (not concave)
Iteration 274: log restricted-likelihood = 704.06629 (not concave)
Iteration 275: log restricted-likelihood = 704.06629 (not concave)
Iteration 276: log restricted-likelihood = 704.06629 (not concave)
Iteration 277: log restricted-likelihood = 704.06629 (not concave)
Iteration 278: log restricted-likelihood = 704.06629 (not concave)
Iteration 279: log restricted-likelihood = 704.06629 (not concave)
Iteration 280: log restricted-likelihood = 704.06629 (not concave)
Iteration 281: log restricted-likelihood = 704.06629 (not concave)
Iteration 282: log restricted-likelihood = 704.06629 (not concave)
Iteration 283: log restricted-likelihood = 704.06629 (not concave)
Iteration 284: log restricted-likelihood = 704.06629 (not concave)
Iteration 285: log restricted-likelihood = 704.06629 (not concave)
Iteration 286: log restricted-likelihood = 704.06629 (not concave)
Iteration 287: log restricted-likelihood = 704.06629 (not concave)
Iteration 288: log restricted-likelihood = 704.06629 (not concave)
Iteration 289: log restricted-likelihood = 704.06629 (not concave)
Iteration 290: log restricted-likelihood = 704.06629 (not concave)
Iteration 291: log restricted-likelihood = 704.06629 (not concave)
Iteration 292: log restricted-likelihood = 704.06629 (not concave)
Iteration 293: log restricted-likelihood = 704.06629 (not concave)
Iteration 294: log restricted-likelihood = 704.06629 (not concave)
Iteration 295: log restricted-likelihood = 704.06629 (not concave)
Iteration 296: log restricted-likelihood = 704.06629 (not concave)
Iteration 297: log restricted-likelihood = 704.06629 (not concave)
Iteration 298: log restricted-likelihood = 704.06629 (not concave)
Iteration 299: log restricted-likelihood = 704.06629 (not concave)
Iteration 300: log restricted-likelihood = 704.06629 (not concave)

```

convergence not achieved

Computing standard errors:

standard-error calculation has failed

Mixed-effects REML regression

Group variable: **Number**

Number of obs = 45

Number of groups = 16

Obs per group:

min = 2

avg = 2.8

max = 3

Wald chi2(0) = .

Prob > chi2 = .

Log restricted-likelihood = 704.06629

V0	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
GroupNumerical						
1	0 (omitted)					
2	0 (omitted)					
TimeNumerical						
30	3.08e-16	6.87e-16	0.45	0.654	-1.04e-15	1.65e-15
74	3.08e-16	6.87e-16	0.45	0.654	-1.04e-15	1.65e-15
GroupNumerical#TimeNumerical						
1 30	-1.85e-15	1.04e-15	-1.77	0.077	-3.89e-15	1.98e-16
1 74	-2.10e-15	1.11e-15	-1.89	0.059	-4.29e-15	8.15e-17
2 30	7.18e-16	9.39e-16	0.76	0.445	-1.12e-15	2.56e-15
2 74	8.20e-16	9.72e-16	0.84	0.398	-1.08e-15	2.72e-15
Sex	0 (omitted)					

weight	0 (omitted)
_cons	0 (omitted)

Random-effects parameters	Estimate	Std. err.	[95% conf. interval]
Number: Identity			
var(_cons)	43.07886	.	.
var(Residual)	1.23e-30	.	.

LR test vs. linear model: chibar2(01) = 1651.46 Prob >= chibar2 = 0.0000

Warning: Convergence not achieved.

Warning: Standard-error calculation failed.

26 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
V0			
TimeNumerical@GroupNumerical			
0	(omitted)		
1	(omitted)		
2	(omitted)		
Joint	(omitted)		

	Contrast	Std. err.	z	P> z	[95% conf. interval]
V0					
TimeNumerical@GroupNumerical					
(30 vs base) 0	3.08e-16	6.87e-16	0.45	0.654	-1.04e-15 1.65e-15
(30 vs base) 1	-1.54e-15	7.85e-16	-1.96	0.050	-3.08e-15 -4.64e-19
(30 vs base) 2	1.03e-15	6.41e-16	1.60	0.109	-2.30e-16 2.28e-15
(74 vs base) 0	3.08e-16	6.87e-16	0.45	0.654	-1.04e-15 1.65e-15
(74 vs base) 1	-1.79e-15	8.77e-16	-2.05	0.041	-3.51e-15 -7.53e-17
(74 vs base) 2	1.13e-15	6.87e-16	1.64	0.101	-2.18e-16 2.47e-15

27 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
v0			
GroupNumerical@TimeNumerical			
0	(omitted)		
30	(omitted)		
74	(omitted)		
Joint	(omitted)		

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
v0						
GroupNumerical@TimeNumerical						
(1 vs base) 0	0	(omitted)				
(1 vs base) 30	-1.85e-15	1.04e-15	-1.77	0.077	-3.89e-15	1.98e-16
(1 vs base) 74	-2.10e-15	1.11e-15	-1.89	0.059	-4.29e-15	8.15e-17
(2 vs base) 0	0	(omitted)				
(2 vs base) 30	7.18e-16	9.39e-16	0.76	0.445	-1.12e-15	2.56e-15
(2 vs base) 74	8.20e-16	9.72e-16	0.84	0.398	-1.08e-15	2.72e-15

```

28 .
29 .
30 .
    end of do-file
31 .

```